Making the Emergency Route System Different from Google Maps

How ML is used, and extra ML features to add - Uncertainty-Aware Emergency Route and ETA Prediction System

1. Where ML is used now

- **Segment travel-time prediction (XGBoost / LightGBM):** a regression model predicts how long each road segment takes. This becomes the edge weight for A*.
- **Uncertainty prediction (quantile regression):** instead of one number, the model predicts a lower guess (10th percentile) and an upper guess (90th percentile). The user sees a range such as 12-18 min.
- **Fastest vs reliable route:** the fastest route runs A* on the typical time. The reliable route runs A* on the 90th-percentile (worst-case) time.
- Pathfinding itself (A*), rerouting and graph building are classical algorithms, not ML.

2. Quantile regression in simple words

- A normal model gives one number, like 15 minutes. Real travel time is never exact, so quantile regression gives a range.
- The lower guess means only 1 in 10 trips is faster. The upper guess means only 1 in 10 trips is slower. About 8 in 10 trips fall between them.
- Example: past times on a road are 10, 12, 13, 15, 15, 16, 18, 25 sec. Normal model says 15. Lower quantile says 11. Upper quantile says 22.
- Both models use the same data and features. Only the training loss (the target) is different.

3. Features Google Maps does not offer

- **Ambulance-specific travel time (strongest idea):** ambulances use sirens, jump signals and get road space. Train on real ambulance trips, or build a correction model that converts car time into ambulance time.
- **Best hospital, not just nearest:** predict the ER waiting time of each hospital (bed availability, specialty, crowd). Choose the hospital with the lowest total time = travel time + wait time.
- **Route by emergency type:** heart or trauma cases need smooth roads. Use road quality, speed breakers, potholes and number of turns as an extra A* objective.
- **Traffic signal and junction delay:** predict waiting time per junction from time of day, junction type and turn direction.
- **Predicting future jams (proactive rerouting):** predict e.g. 'in 10 min this road has a 70% chance of a jam' and reroute before the jam happens.
- **Weather and events:** rain, fog, school timings, matches, festivals, processions. Use SHAP or feature importance to explain why the ETA changed.
- **Confidence-based decisions:** Google shows one number. This project shows a range plus confidence, and suggests a safer route when the range is very wide.

4. Recommendation for a final-year project

- Pick these three, because ML does real work in them: (1) hospital total time = travel + wait, (2) ambulance correction model, (3) prediction of upcoming traffic for proactive rerouting.

- Add road quality and signal delay later if time permits.

5. Evaluation

- Report MAE and RMSE for the central prediction.
- Also report coverage: how often the real time falls inside the predicted range. A 10th-90th range should cover about 80% of trips.

6. Risks and open points

- **Data is the main risk.** Hospital wait time and real ambulance trips are hard to get publicly. Where data is missing, use synthetic data and say so clearly in the report.
- Adding per-segment quantiles to get a route range overstates the width. Summing is acceptable for a student project if the limitation is stated.
- Open decisions: definition of 'reliable' (worst-case 90th percentile is recommended), update frequency (30-60 s), alert threshold, and the A* heuristic (straight-line distance / max speed).
- This document is based only on the two project PDFs. No dataset was checked.